

Madhurima Munagala

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PROFESSIONAL SUMMARY

Computer Science undergraduate at VIT Vellore with experience in full-stack development, software engineering, and AI applications using Java, Python, React, FastAPI, and PostgreSQL. Published researcher and IBM Certified Generative AI Engineer passionate about building scalable software solutions.

EDUCATION

Vellore Institute of Technology

B.Tech Computer Science and Engineering
CGPA: 8.96

Vellore, India
August, 2023 - July, 2027

Narayana Junior College

Higher Secondary School | Score: 97.3%

Tirupati, Andhra Pradesh
May, 2023

CERTIFICATION

IBM Certified Generative AI Engineer

IBM

- Strengthened expertise in Large Language Models, prompt engineering, transformer architectures, and responsible AI through hands-on implementation.
- Built AI-powered applications using Python, Streamlit, and NLP for document analysis and workflow automation.

PUBLICATIONS

Smartphone-Based Real-Time Crowd Monitoring System for Smart City Applications

Apr 2026

SSRN (Elsevier)

- Sole author of the SSRN (Elsevier) publication "*Smartphone-Based Real-Time Crowd Monitoring System for Smart City Applications*", proposing a smartphone-based crowd monitoring framework for smart city applications.
- Presented the system architecture, Android implementation using Camera2 API, and a scalable YOLO + TensorFlow Lite pipeline for real-time on-device crowd estimation and future AI-based person detection.
- https://papers.ssrn.com/sol3/papers.cfm?abstract_id=6573558

PROJECTS

Smart Grocery Expiry & Waste Reduction System | React, FastAPI, Python, PostgreSQL

- Developed a full-stack grocery management system with OCR-based receipt parsing, expiry tracking, RESTful APIs, recipe recommendation, and PostgreSQL integration, reducing manual tracking effort by 60% and improving query performance by 35%.
- <https://fridgeproj.vercel.app/>

Movie Night Matcher | React, Firebase, Firestore, TMDB API

- Built a real-time collaborative movie matching platform using React, Firebase Authentication, Cloud Firestore, and TMDB API, enabling friends to create/join rooms and vote on movies together.
- Implemented synchronized multi-user voting with Firestore real-time listeners and server-side Cloud Functions to detect group matches, while integrating TMDB for movie details, ratings, posters, and streaming availability.
- <https://movie-night-matcher-63275.web.app/>

Resume Screening Assistant | Python, Streamlit, pdfplumber, Pandas, NLP, Regular Expressions

- Developed a Resume Screening Assistant using Python, Streamlit, and pdfplumber to classify PDF resumes into job roles using NLP techniques.
- Implemented multi-file PDF processing, interactive result visualization, and CSV report generation, providing a lightweight solution for automated resume screening.

TECHNICAL SKILLS

Languages: Java, Python, C++, C

Technologies: React, HTML, CSS, Tailwind CSS, FastAPI, Flask, Node.js, REST APIs, Firebase

AI/ML: YOLO, TensorFlow Lite, OpenCV, Pandas, NumPy, Matplotlib

Databases & Tools: PostgreSQL, MongoDB, Firebase Firestore, Git, GitHub, VS Code, IntelliJ IDEA, PyCharm, Android Studio, StarUML, Modelio, ProjectLibre, Figma